2nd Order DEs: Introducing Damping

Questions 1–14

Question 1. The following equations are each the solution of a different homogeneous second order linear differential equation with constant coefficients. In each case, determine whether the solution shows that the differential equation was modelling heavy damping (overdamping), critical damping, light damping (underdamping), or neither.

a) $x = 4e^{-t} + 2e^{-2t}$
b) $x = e^{-3t}(4\cos t - 3\sin t)$
c) $x = 7te^{-4t} + 4e^{-4t}$
d) $x = 2e^{-6t} + 5e^{3t}$
e) $x = e^{4t}(5\cos 2t + 8\sin 2t)$
f) $x = 9e^{-t} - 3te^{-t}$
g) $x = e^{-7t}\left(3\sin\frac{t}{2} - 4\cos\frac{t}{2}\right)$
h) $x = 7e^{-6t} - 10e^{-12t}$
i) $x = 8te^{t} - 3e^{t}$
j) $x = 5e^{-4t}\cos 6t$

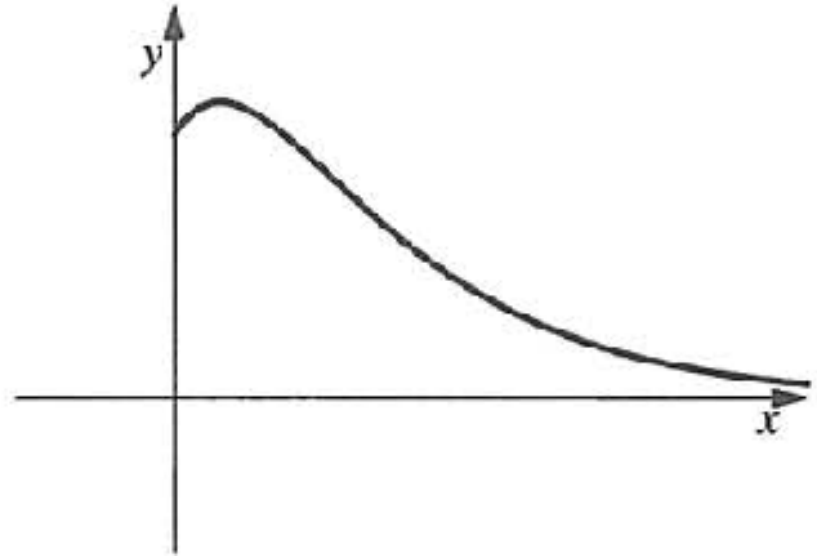
Question 2. Solve the following differential equations, classifying each scenario as heavy damping (overdamping), critical damping, or light damping (underdamping).

a) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 5y = 0$
b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$
c) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$
d) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$
e) $\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 30y = 0$
f) $2\frac{d^2y}{dx^2} + 7\frac{dy}{dx} + 5y = 0$
g) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$
h) $2\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 17y = 0$
i) $9\frac{d^2y}{dx^2} + 30\frac{dy}{dx} + 25y = 0$

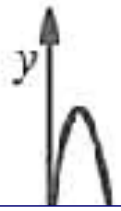
Question 3. The following graphs show solutions to differential equations of the form

$$\frac{d^2y}{dx^2} + k\frac{dy}{dx} + \omega^2 y = 0,$$

where $k^2 > 4\omega^2$ or $k^2 < 4\omega^2$. Describe what is happening in each scenario.



a)



Question 4. By finding the coordinates of any stationary points first, sketch the following graphs, given that $x \geq 0$.

a) $y = 3e^{-x} - 2e^{-2x}$
b) $y = 4e^{-3x} + 3e^{-x}$
c) $y = -3e^{-2x} - 6e^{-4x}$
d) $y = (2x - 1)e^{-2x}$

Question 5. Sketch the following graphs, given that $x \geq 0$.

a) $y = e^{-\frac{1}{2}x} \cos x$

b) $y = e^{-\frac{1}{3}x} \sin 3x$

c) $y = e^{-x}(\cos 2x + 2 \sin 2x)$

d) $y = e^{-3x}(-3 \cos 4x + \sin 4x)$

Question 6. It is given that $2\ddot{x} + 5\dot{x} + 3x = 0$, $t \geq 0$, where $x = 2$ and $\dot{x} = 1$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) Find the value of t when $\dot{x} = 0$.

d) Sketch the graph of x against t.

Question 7. It is given that $4\ddot{x} + 4\dot{x} + 5x = 0$, $t \geq 0$, where $x = 2$ and $\dot{x} = -1$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) State the period of the oscillations.

d) Sketch the graph of x against t.

Question 8. By finding the coordinates of any stationary points first, sketch the following graphs, given that $x \geq 0$.

a) $y = 4e^{-3x} - 2e^{-2x} + 1$

b) $y = e^{-2x} - 2e^{-3x} - 2$

c) $y = 3e^{-x} - 3e^{-2x} - 1$

d) $y = (3 - 9x)e^{-3x} + 2$

Question 9. Sketch the following graphs, given that $x \geq 0$.

a) $y = 3e^{-\frac{1}{2}x}\cos 2x + 3$

b) $y = e^{-\frac{1}{4}x}\sin x - 1$

c) $y = e^{-2x}(3\cos 4x - 2\sin 4x) - 3$

d) $y = e^{-x}(\cos 3x + 11\sin 3x) - 2$

Question 10. It is given that $\ddot{x} + 6\dot{x} + 8x = 8$, $t \geq 0$, where $x = 2$ and $\dot{x} = 2$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) Find the value of t when $\dot{x} = 0$.

d) Sketch the graph of x against t.

Question 11. It is given that $16\ddot{x} + 8\dot{x} + x = -3$, $t \geq 0$, where $x = 1$ and $\dot{x} = 1$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) Find the value of t when $\dot{x} = 0$.

d) Sketch the graph of x against t.

Question 12. It is given that $4\ddot{x} + 8\dot{x} + 13x = 26$, $t \geq 0$, where $x = 1$ and $\dot{x} = 4$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) State the period of the oscillations.

d) Sketch the graph of x against t.

Question 13. It is given that

$$\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 17y = 40\cos x, \qquad x \geq 0,$$

where $y = 2$ and $\frac{dy}{dx} = 2$ when $x = 0$.

a) Find a particular solution to the differential equation.

b) Describe what happens when x is large.

Question 14. It is given that

$$4\frac{d^2y}{dx^2} + 9y = \frac{9}{2}x, \qquad x \geq 0,$$

where $y = 4$ and $\frac{dy}{dx} = -1$ when $x = 0$.

a) Show that a particular solution to the differential equation is $y = 4\cos\frac{3}{2}x - \sin\frac{3}{2}x + \frac{1}{2}x$.

b) Find the smallest value of x such that y is stationary, giving your answer to three significant figures.

c) Sketch the graph of y against x.

The differential equation is modified so that the curve continues to oscillate about $y = \frac{1}{2}x$ but with reducing amplitude. The solution is now

$$y = e^{kx}\left(4\cos\frac{3}{2}x - \sin\frac{3}{2}x\right) + \frac{1}{2}x.$$

d) Explain why $k < 0$.

e) Show that $\left.\frac{dy}{dx}\right|_{x=0} \neq -1$.

f) Sketch the graph of y against x.